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LUMI: Data-Driven Environmental Intelligence System for Renewable Energy Decision Support

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CHAPTER 1

THE PROBLEM AND REVIEW OF LITERATURE

This chapter provides context to the background of study, problem statement, objectives, scope, and the beneficiaries of the study. Related literature is also discussed in the section.

1.1 PROBLEM AND ITS BACKGROUND

1.1.1 Project Context

Climate change in the Philippines is intensifying, leading to frequent extreme weather events such as heat waves, strong typhoons, and heavy rainfall. The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) indicates that these climate shifts since the mid-20th century are primarily driven by human activities, particularly increased greenhouse gas emissions from land-use changes and fossil fuel combustion. These factors exert mounting pressure on the environment and society, with future implications anticipated as the country remains heavily reliant on fossil fuels for energy, underscoring the urgent need for renewable and sustainable energy solutions.

Lenain (2026) highlighted that the Philippine government is making significant investments in climate adaptation initiatives to mitigate the long-term effects of climate change. Additionally, the country is enhancing its renewable energy contributions as part of global climate mitigation efforts. Nevertheless, the adoption



of renewable energy faces challenges, particularly due to fragmented climate and energy-related data that are often presented in complex formats, hindering public understanding and utilization. Although information on climate change is abundant, there is a noticeable scarcity of user-friendly, data-driven tools that effectively illustrate the consequences of fossil fuel use and the feasibility of transitioning to renewable energy sources.

There is a significant amount of climate and energy-related data available, however, there is a lack of systems that effectively transform this data into accessible and actionable insights for users. Existing information sources often present data in ways that are difficult to interpret, limiting the ability of individuals to make informed decisions regarding energy use and sustainability.

To address this gap, this study proposes Lumi, a data-driven environmental intelligence system designed to provide users with accessible insights and predictive analytics. It aims to support users in understanding current energy consumption and demand trends and evaluating suitable renewable energy options through data visualizations, scenario-based simulations, and recommendation mechanisms which includes the costing needed.



1.2 PROBLEM STATEMENT

Despite the growing awareness of the public about renewable energy, it is still poorly understood and seen as an expensive investment to some. With a significant portion of the Philippines' electrical grid still relying on fossil fuels, there is a growing need to transition and increase the country's energy output to match current demand. The lack of a system that allows the public to be educated and evaluate their renewable energy source options will result in the country still relying on fossil fuels. This study aims to tackle this issue to educate users and allow them to have informed decisions when choosing the appropriate renewable energy source for their application.

1.2.1 HYPOTHESIS

1.2.1.1 Null Hypothesis

LUMI does not demonstrate a significant level of effectiveness in accurately providing data-driven insights for understanding and evaluating renewable energy options.

1.2.1.2 Alternative Hypothesis

LUMI demonstrates a significant level of effectiveness in accurately providing data-driven insights for understanding and evaluating renewable energy options.



1.3 STATEMENT OF OBJECTIVES

1.3.1 General Objective

To develop a web-based environmental intelligence system that provides accessible, data-driven insights and predictive analytics to help users understand and evaluate renewable energy options.

1.3.2 Specific Objectives

1.3.2.1 To design and develop a dashboard that visualizes climate patterns and energy trends interpretation.

1.3.2.2 To develop a rule-based recommendation engine for identifying suitable renewable energy sources.

1.3.2.3 To implement a predictive analytics module for forecasting energy demand trends.

1.3.2.4 To create a simple what-if simulation tool for comparing renewable energy scenarios.

1.3.2.5 To develop a user-friendly interface that supports data interpretation and decision-making.

1.3.2.6 To evaluate the usability, usefulness, and acceptability of the proposed renewable energy recommendation system among potential users.

1.3.2.7 To test the system performance metrics including the accuracy, recall, F1-score, response time, and memory utilization



1.4 SCOPE AND DELIMITATIONS OF THE STUDY

1.4.1 Scope of the Study

This study evaluates Lumi, a web-based environmental intelligence system designed to assist users in assessing renewable energy options. It combines publicly available climate and energy data, like temperature and rainfall patterns with interactive visualizations. Key features include a climate and energy dashboard, predictive analytics for energy demand, a rule-based recommendation engine for renewable sources (solar, wind, hydro, and geothermal), and a what-if simulation tool. These tools in the system will use various modules such as the energy hub dashboard, eco-simulator, and the forecasting module. Additionally, it offers a region selection module for localized insights within the Philippines. Lumi targets communities and general users, focusing on user-friendly outputs to improve decision-making in renewable energy selection.

1.4.2 Limitations of the Study

The study has several limitations: it relies on publicly available datasets, affecting the accuracy of results. The predictive analytics module uses basic statistical methods instead of advanced machine learning, to improve computational costs. The recommendation engine is rule-based and will focus on criteria set in the system. It focuses on specific renewable sources such as solar, wind, and hydro, excluding non-renewable technologies such as fossil fuel based sources. Additionally, the what-if



simulation tool simplifies comparisons for better user comprehension. It will also provide initial cost estimates for implementing renewable energy sources. The system serves to provide information to the user and improve their comprehension of information related to renewable energy sources.

1.5 SIGNIFICANCE OF THE STUDY

For students and educators, it serves as an educational resource that facilitates learning about climate change and renewable energy technologies. For households and community members, it offers practical insights to promote awareness and foster the adoption of sustainable energy practices.

Household residents and homeowners can utilize the system as a decision-support tool to grasp concepts related to renewable energy technologies and energy production. By organizing and interpreting data effectively, the system aids in crafting informed strategies for promoting renewable energy, initiating sustainability initiatives, and improving perception towards renewable energy.

Communities as well as government institutions such as the Department of Energy, Department of Environment and Natural Resources, the National Grid Corporation of the Philippines, and local government units can use the system to plan the adoption of renewable energy sources in the country. These institutions will be



able to create policies that suit the needs of their localities by providing them with data-driven insights to allow them to make informed decisions.

For researchers and developers, the study lays a groundwork for creating advanced environmental intelligence systems that incorporate artificial intelligence and data analytics. It showcases the practical application of rule-based AI and predictive analytics in tackling real-world environmental challenges.

1.6 REVIEW OF RELATED LITERATURE AND STUDIES

Studying adoption of renewable energy as well as factors that affect its adoption rates is crucial to understanding the needs of the users of LUMI and the sectors it aims to serve.

Renewable energy adoption

There have been a variety of factors that affect the adoption of renewable energy. However, these factors vary based on location and the economy of the country. Khalid et al. (2021), evaluated the adoption of renewable energy by consumers in Poland, a developed country from Europe. Initial costs and risk are seen as the least influencing factors, while environmental concerns, ease of use, and financial incentives are seen as the key influencing factors in adopting renewable sources in Poland (Khalid et al., 2021). Regardless of the purpose of the energy source, perceptions and preference can affect the transition to a new energy source.



Malaysia, a Southeast Asian country in the region, is encountering a lower than the target percentage renewable energy source in their electricity grid, the study revealed that attitude and control over the energy source are seen as major factors that affect intentions to use renewable energy (Wong et al., 2023). Preferences and perceptions of different energy sources can significantly affect the people's decision to transition, and education can further affect their perception (Aguilera et al., 2024).

Public perception also affects policies that govern energy transition, and public engagement as well as increasing the awareness of the public in climate change can shift their sentiment in favor of renewable energy sources (Esiri et al., 2024). Moreover, the type of renewable energy can also affect a person's decision to transition and if they can hire external labor to execute the transition for them (Danne et al., 2021). In developing countries such as the Philippines, financial limitations, governance challenges, and societal resistance to renewable energy have been the barriers to renewable energy adoption (Zhindon-Almeida & Ruiz-Carrillo, 2025). Another study conducted by Rana et al. (2025), stated that these barriers to adopting renewable energy significantly affects intentions to adopt renewable energy particularly in developing countries compared to developed counterparts. The Philippine grid has a capacity of around 27.0GW of energy output, it is heavily dependent on fossil fuels which makes up for around 70.5% of the country's energy production, 43.1% of which is Coal and 29.5% of energy generated in the country came from renewables (Gonocruz et al., 2024). A study conducted by Palanca-Tan et



al. (2024), stated that there is a greater preference to solar energy as a renewable energy source for electricity, and that Filipinos in Metro Manila are more willing spend more in solar energy over biomass, wind, and small-scale hydropower indicating a strong preference to a particular type of renewable energy. In rural areas in the Philippines such as island communities, various community-based renewable energy initiatives have been established to provide access to electricity via renewable energy to different communities, such projects have provided a positive impact to the island communities they serve (Franco & Taeihagh, 2024).

The studies above have shown that there are a variety of factors that affect the adoption of renewable energy sources. Environmental concerns and financial incentives are seen as key drivers of adoption in developed, first-world countries such as Poland. However, public perception, attitude, perceived cost, and education still affect the intention to adopt renewable energy. In the Philippines, only 29.5% of the electric grid is powered by renewable energy and still has room for expansion. With community-based renewable energy, the adoption rate and access to electricity produced from renewable energy is also increasing in island communities. There is a strong preference for solar power by Filipinos in Metro Manila which might however, there is still need for additional data beyond Metro Manila. Educating individuals about renewable energy and their options can improve adoption in the country.

Environmental Intelligence



Environmental intelligence as defined by Bassetti (2024), is the use of machine learning algorithms and AI to simulate predicted environment models. AI has been used as a part of environmental intelligence to help curb the amount of CO₂ emissions, by providing estimations on renewable energy consumption and renewable energy optimization (Khan et al., 2025). Forecasting models have been used for determining the power generation with renewable energy, particularly with solar and wind. A study conducted by Das et al. (2022), have used a variety of algorithms such as support vector regression (SVR) and artificial neural network to forecast solar power output using specific meteorological data such as solar irradiance and atmospheric temperature, and forecasted output can be improved further with optimized SVR-parameters. Long Short-Term Memory deep learning models can also be used to create prediction models for solar irradiance by utilizing known correlations between weather parameters and solar irradiance which can also provide data for forecasting solar energy output (Bandara et al., 2026). A hybrid approach in predicting wind power output based on deep auto-encoding convolutional neural network long short-term memory model to allow for effective modelling of offshore wind power (Khan et al., 2021).

Forecasting models for predicting source load for renewable energy demand have been created using conditional generative adversarial nets and multi-gate mixture-of-experts (Wang et al., 2025). Such models are useful in anticipating electricity demand and allow data-driven decisions for managing renewable energy



outputs. According to Deming et al. (2021), technologies such as renewable energy scheduling can allow algorithms to better maximize renewable energy output and energy storage deployment, incorporation of these strategies is necessary to achieving stable renewable energy.

AI and machine learning can be used to mitigate the intermittency of renewable energy such as solar and wind to maximize generation and storage (Ojuekaiye, 2025). According to the study of Algburi et al. (2025), AI can effectively optimize renewable energy by predicting supply and demand while also enabling decentralized energy integration which are closer to the consumer side of the electric grid. Such systems that integrate environmental data and forecasting models can be integrated to create a robust data-driven model for predicting renewable energy output. This was discussed in an article by Hannan et al. (2021), which highlighted that while AI can improve the efficiency of the renewable energy system in its role in managing the utilization of renewable energy systems. A study conducted by Benitez et al. (2024), realized that current forecasting models for the renewable energy sources in the Philippines have been found to be volatile and fails to meet the Mean Absolute Percentage Error (MAPE) threshold of 18% set by the Philippines' Wholesale Electricity Spot Market (WESM), such findings indicate to have improved data-driven forecasting models.

The use of environmental intelligence to forecast energy usage and output is necessary to create an optimized renewable energy system that meets demand.



Meteorological data is seen as an important source of data for predicting energy output, especially for solar and wind that rely on solar irradiance and windspeed, respectively. The use of hybrid algorithms are also present in the studies above such as the use of DACNN-LSTM to model offshore power. Currently, one of the key concerns in renewable energy systems is that they experience intermittency that can be handled with the right forecasting models and energy infrastructure. Furthermore, current studies in forecasting models for the renewable energy sources in the Philippines have demonstrated the unreliability for current systems in place and must be improved.

Multi Criteria Decision Making (MCDM)

There have been a number of researches that used Multi Criteria Decision Making to support ongoing efforts to utilize and transition to renewable energy sources. The preference ranking organization method for enrichment evaluations (PROMETHEE) is a type of MCDM that compares a limited set of options using performance scores of different criteria to determine which option will outrank others, a study conducted that utilized a modified PROMETHEE approach that integrates multi period multi-attribute decision making, determined that it can aid in improving the comprehension of various problems by structuring it into a comprehensive evaluation (Witt & Klumpp, 2021).



Using various methods such as Analytical Network Process, Fuzzy Logic, and Weighted Linear Combination, a study have managed to proposed four renewable energy sites in Iran by allocating 57% of the entire study area to build a variety of renewable energy sites such as biomass, geothermal, solar, and wind power plants (Shorabeh et al., 2021). Wind power plants, a type of renewable energy source, requires a suitable location in order to reliable energy generation, using various MCDM methods such as AHP, PROMETHEE, and FAHP and combining them results in an accurate site selection by ensuring objectivity with decision-making (Abdullah et al., 2025). A study conducted by Estévez et al. (2021), sought to improve the participation of locals on topics concerning decision-making related to renewable energy, results showed that multi-criteria decision analysis (MCDA) had improved social impact but still needed expert consultations.

The type of renewable energy most appropriate to use varies from different regions as different countries will have different accessibility to each type and must align in the economics of the region (Shatnawi et al., 2021). It is identified that in a criteria based decision making that technical criterion holds the most weight at decision making at 53.6%, followed by environmental impact at 29.0% meanwhile, geographical location and socioeconomic criteria are the lowest at 11.3% and 6.0%, respectively (Shatnawi et al., 2021). Indecision can often impact decision-making and MCDA is seen as an effective aid, a study that was conducted by Ezbakhe and Pérez-Foguet (2021), modified the EElimination and Choice Translating REality



(ELECTRE) III model to curb uncertainty and facilitate informed decision making to evaluate various renewable options for Turkey, using Turkey's Vision 2023 targets, wind and solar has been identified as the most appropriate renewable energy source. A study conducted by Mandal et al. (2025), utilized Weighted Aggregated Sum Product Assessment and FANMA to handle various decision-making scenarios to identify and rank renewable energy options for Serbia.

In the Philippines, local studies have been conducted that utilized MCDM and MCDA for decision making and analysis. A study conducted by Ongpeng et al. (2021), focused on applying MCDA framework to retrofit buildings to integrate renewable energy sources, using a hybrid Analytic Hierarchy Process and VIKOR, this approach allowed engineers to support retrofitting decisions while meeting performance dimensions.

Potential sites for single as well as hybrid renewable energy sources that uses various renewable energy sources such as hydropower, wind, and solar have been studied to identify their feasibility to developers, certain factors such as socio-environmental and techno-economic factors have been identified to affect the suitability of sites for renewable energy systems (Tarife et al., 2023). Machine learning models that uses the MCDM method have been used to facilitate decision making that takes into account the feasibility and profitability in identifying optimal locations for wind power plants in the Philippines, which provides the policymakers,



investors, and other stakeholders to make informed decisions involving investment to the renewal energy sector (Cerna et al., 2023).

Studies from foreign and local sources provided a variety of options for MCDM systems. PROMETHEE used ranking based on performance while ELECTRE can aid in reducing indecision among decision makers. A number of studies have also used hybrid systems that combine multiple algorithms to maximize accuracy in providing decision support to the users, often combining at least two to three algorithms. The study would benefit in using a combination of multiple algorithms to maximize the accuracy of the system and would benefit the beneficiaries of this study.

Intelligent decision support systems (IDSS)

A data driven decision support system requires a variety of useful data to provide the most accurate decision and recommendation. A study conducted by Krishnan et al. (2024), utilized a number of algorithms such as AutoRegressive Integrated Moving Average (ARIMA) with Reinforcement Learning (RL) to predict energy demand and allocate resources dynamically and utilized Internet of Things (IoT) devices as a source of data to help reduce wasted energy. In forecasting and fault detection, neural networks such as convolutional neural network, recurrent neural network, long short term memory, and graph neural networks, with hybrid models that combines deep learning and optimization algorithms provide improved



accuracy in system design and energy management (Hu et al., 2026). A study by Gonçalves et al. (2025), created a Machine Learning as a Service (MLaaS) framework to create renewal energy microgrids that leveraged a RL and an open-source simulator Pymgrid to forecast electricity prices and decision making in utilizing energy sources from both the renewable energy microgrid community and the main grid.

A study in Bahrain utilized a hybrid analytical hierarchy process with artificial neural networks to determine the suitable renewable energy source for the country and determined that a hybrid system allowed it to generate scenarios and accurately forecast results (Alabbasi et al., 2024). A web-based decision support system in the United Kingdom was made to determine the feasibility of onshore renewable energy sources utilized an MCDA method specifically, the weighted sum model (WSM) to assess, score and rank the strengths and weaknesses of renewable energy options, the lack of need for pairwise comparison allowed the model to run assess the feasibility of renewable energy infrastructure faster (Beriro et al., 2022). Decision support system studies have used a number of different technologies that are designed to improve the efficacy of decision provided by these systems. A temporal MCDA decision support system that utilized the Data vARIability Assessment- Evaluation based on Distance from Average Solution (DARIA-EDAS) method have provided an automated, rapid, objective assessment of problems that required temporal decision



making to produce unambiguous results that can be interpreted easily (Bączkiewicz et al., 2024).

In a study by Panagoulas et al. (2023), examined the use of explainable artificial intelligence (XAI) in intelligent decision support systems and concluded that it is part of the main feature of IDSS and that it ensures that it is tailored to the needs of the user and that to increase personalization of such systems, the users must be clustered by their perception of usefulness and ease of use of the system. Production-distribution systems with energy awareness for renewable energy have also been integrated with decision support systems to aid in reducing CO2 emissions and improve company profits, these systems are capable of assessing and planning under various circumstances (Mohammadpour & Rahmati, 2023). Intelligent decision support systems have been used to provide decision support for farms powered with renewable energy, using a layered approach combining sensors and machine learning in predicting energy demand with coordination with varying working conditions to greater efficiency, reduced downtime, and continuous operations (Adelakun et al., 2025).

Decision support systems that integrated machine learning and AI is seen as an effective in handling various challenges related to incomplete data, scalability, and generalizability, in the context of retrofitting buildings to accommodate renewable energy systems, these systems have enhanced the capabilities of retrofitting such as with accuracy and reliability of energy performance predictions (Baset & Jradi,



2024). Various studies have shown that in various scenarios in implementing and improving renewable energy deployment, intelligent decision support systems have provided improvements in decision making and have allowed for data-driven decisions to be taken in these different scenarios about renewable energy.

Various studies have shown that in various scenarios in implementing and improving renewable energy deployment, intelligent decision support systems have provided improvements in decision making and have allowed for data-driven decisions to be taken in these different scenarios about renewable energy. The use of artificial intelligence and machine learning to provide an intelligent decision support system is feasible and have been used a number times. Algorithms such as ARIMA with RL can provide the system with a dynamic decision making capability that is capable of adapting with most scenarios. Meanwhile, XAI systems can provide users with information that is unambiguous that also adapts with the users needs and preferences.

Predictive environmental modeling

Predictive environmental modeling can be useful in determining energy demand that could then shape the designed output of renewable energy systems. A study conducted by Aljohani and Almansour (2025), utilized predictive models to forecast energy demand using the Renewable energy generation, energy consumption, prices, and weather in Spain (2015–2018) dataset, using CNN-LSTM for forecasting



and particle swarm optimization (PSO), the model managed to provide strong prediction performance for solar, wind, and hydro power. A study in China utilized a random forest model to investigate the various factors that affect solar energy production in Tengger Desert Solar Park and to provide insights in optimizing solar energy output and integration into the energy grid, with solar irradiance and temperature being key variables that affect output, and the random forest model used in the study demonstrated good predictive ability during training and testing of the model (Li et al., 2026). Traditional and neural network based statistical models have their own advantages, traditional statistical models are often faster and interpretable baselines, while DNN and hybrid models can identify complex patterns that allow them to produce results with greater accuracy (Malakouti, 2026). A comparison of classical, grey, fuzzy, and intelligent models in predicting energy consumption and generation have been conducted, classical models are sufficient on predictable seasonal data, gray and fuzzy methods are useful for scenarios when environmental data is scarce or have erratic behavior, while intelligent systems such as the Prophet are capable for short and medium term forecast with heterogenous localities (Mustafa & Al-Yozbaky, 2025).

Using a data-driven Model Predictive Control to allocate the flow of power from various renewable energy sources, the usage of such sources can increase the seasonal utilization rate of renewables by up to 19.71% using a multi-energy source configuration (Zhang & Cai, 2025). Apart from random forest regressors, other



machine learning models such as gradient boosting regressor, support vector regression, and neural networks have been utilized in the study conducted by Markelov (2025), the models produced accurate forecasts which had the potential to reduce the diesel generator usage and energy waste from simulated scenarios from historical data. Predicting the cleaning needs of solar panels have also been studied and modeled using various time-series approaches such as LSTM to predict performance, the models highlighted factors that affected the frequency for maintenance needed to maximize efficiency and sustainability of solar systems (Abuzaid et al., 2024).

In smart grid systems, a hybrid machine learning model that utilized LightGBM, XGBoost, LSTM and SHAP have been used to predict model energy consumption in relation to temperature data and building types, which provides insights into how environmental factors affect energy demands and shape energy consumption patterns (Anonto et al., 2025). A study evaluated deep-learning models to determine accuracy in forecasting city energy demand and solar power supply loads that is relevant to ensuring energy stability, using models such as bidirectional LSTM, gated recurrent unit, and temporal convolutional model, with bidirectional LSTM providing superior performance in particular (Alhayd & Todeschini, 2022). A hyper-heuristic algorithm for improving electricity forecasting models using genetic algorithm (GA) for high-level strategy with PSO and Bat algorithm for low level



strategy, the algorithm demonstrated high performance in forecasting electricity demand at the cost of higher computational cost (Cao et al., 2025).

Multiple studies have utilized and favored the use of LSTM for its various advantages, particularly its accuracy in forecasting data. However, the availability of data for usage can also determine the algorithm that can be used to avoid overfitting. Traditional statistical systems and neural network based systems have their own use cases based on the availability of data. For most use cases, classical systems can be used when the environment and scenarios are simple and predictable. Meanwhile, fuzzy methods are better for more erratic scenarios. Furthermore, the complexity of the algorithm that would be used will often lead to higher computational cost. Therefore, the balance of speed and accuracy must be achieved in choosing the appropriate algorithm for predictive modeling.

Definition of Terms

The following terms have been defined to facilitate the understanding of the paper:

Artificial Intelligence (AI) is defined as the use of a computer system to perform tasks that require human intelligence (Copeland, 2026).

Climate Change is the change in the climate of the Earth. Which are a result of a variety of factors such as changes in the atmosphere, geological, chemical, biological, and geographical factors occurring within the planet (Jackson, 2026).



Climate Data are quantitative and qualitative data about the climate and the variables affecting it such as temperature and humidity that have been collected systematically across various locations and periods of time (Climatic Data, n.d.).

Dashboard is a report that uses graphical visualizations to show various data relevant to a person, group, organization, etc (“Dashboard,” 2026).

Data Analytics is a process of looking at raw data using different methods and techniques to extract useful information (The Investopedia Team, 2026).

Data Visualization is defined as the representation of data using graphics such as charts, graphs, maps, etc to make information and patterns from the data accessible (What Is Data Visualization? Definition, Examples, and Learning Resources, n.d.).

Energy data are quantitative measurements of data related to energy such as consumption, demand, and production (What Is Energy Data and How to Use It Smartly? | Insight & Savings, 2025).

Fossil fuels are a type of energy source that uses

Predictive Analytics is a branch of advanced analytics that uses historical data and analytical techniques such as statistical modeling, data mining techniques and machine learning to make predictions about future outcomes (Finn & Smalley, n.d.).

Renewable energy is a type energy source that are harnessed from natural sources that replenishes faster than it is being consumed while also producing far lower emissions of greenhouse gases which includes biomass or biomass, geothermal, hydropower, ocean, solar, and wind energy (United Nations, n.d.).



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